flow of trucks if you fill up every square meter of the road with trucks, cars, and bikes. The traffic moves in the fastest way when there is enough space between vehicles to enable them to travel near the speed limit. It is the same with the teams.

What can we do to make data mesh teams performant? First, we should plan our technology stack to be simple compared to the capabilities of the team. Second, we should embrace platform thinking so as not to force teams to reinvent the wheel every time with yet another monitoring and logging solution. Doing both will allow the team to entirely focus on what is essential—its business domain. To further focus in and not overload the team, we decompose large domains into small, focused domain-oriented data products.

All of the data mesh principles are influenced by socio-technical ways of thinking. Domain-oriented decentralized data ownership and architecture reduce the load by decentralizing responsibility for domains and corresponding data.

Data as a product reduces collaboration between teams to access data by making data findable, accessible, interoperable, and reusable. It limits the load by exposing data in an expected and known format and ports by consuming teams. Self-serve data infrastructure as a platform reduces the load by providing a self-service platform.

Federated computational governance makes collaboration between teams easier by enforcing common standards and patterns. It enables team members' transfers by reducing the possible technology stack within the company.

As you can see, at the heart of every principle is a socio-technical way of thinking. Because of the socio-technical nature of this paradigm shift, it's important to be clear about the benefits that can result from this shift. While data mesh is a socio-technical solution to many problems, some major challenges are associated with it. Let's explore them in detail.

1.7 Data mesh challenges